



FONTYS UNIVERSITY OF APPLIED SCIENCES

Dataspaces Project

Team **Pink Panther**

Dataspace Use-Case Validation Platform

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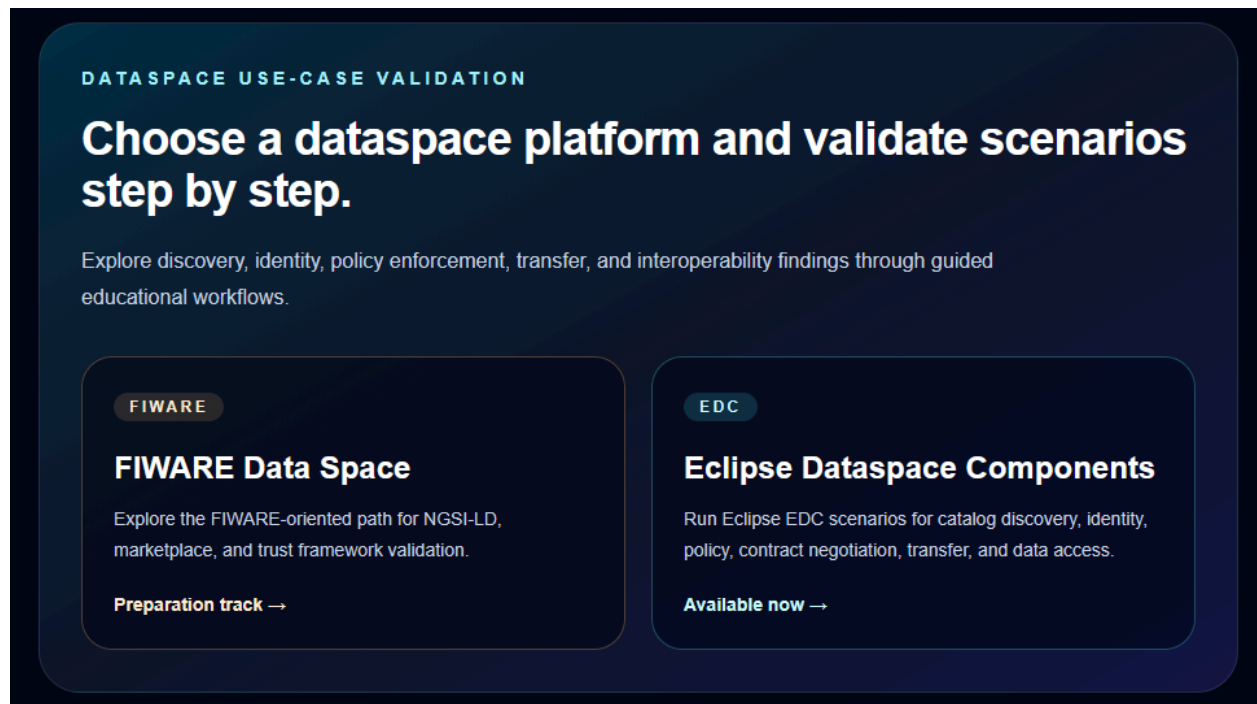


Image 1: Dataspace Use-Case Validation Platform Main Screen

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Version Control

<i>Version</i>	<i>Date</i>	<i>Author</i>	<i>Amendments</i>	<i>Status</i>
<i>1</i>	<i>02.06</i>	<i>Dimitar & Georgi</i>	<i>Initial creation</i>	<i>Done</i>

Feedback Control

Summary

The Dataspace Use-Case Validation Platform is a stakeholder-focused educational demonstrator built on top of the Eclipse Dataspace Components (EDC) Minimum Viable Dataspace (MVD).

The platform provides a visual and traceable environment for validating dataspace interactions such as:

- Catalog discovery
- Contract negotiation
- Policy enforcement
- Transfer initiation
- Data retrieval
- End-to-end dataspace workflows

Instead of requiring users to interact directly with EDC APIs, Kubernetes logs, and debugging tools, the platform presents the complete process through guided scenarios, visual workflows, and execution traces.

The dashboard was originally created as a debugging companion for the Eclipse MVD and later evolved into a Dataspace Use-Case Validation Platform following stakeholder feedback requesting:

- better usability
- less technical complexity
- stronger educational value
- support for future FIWARE and EDC integration scenarios

Introduction

Problem Statement

While the Eclipse MVD successfully demonstrates dataspace interactions, understanding the execution flow is difficult for:

- students
- project stakeholders
- non-technical contributors
- future researchers

Critical information is distributed across:

- EDC control planes
- EDC data planes
- browser requests
- Kubernetes resources
- vault services
- IdentityHub services

This makes validation and troubleshooting difficult.

Proposed Solution

Develop a unified validation platform that:

- executes dataspace workflows
- visualizes each interaction
- stores execution traces
- simplifies stakeholder demonstrations
- supports educational use cases

Requirements

Stakeholder Requirements

Stakeholder feedback identified several important requirements.

- Requirement 1

The platform must focus on use cases rather than raw technical implementation.

- Requirement 2

The platform must provide visual guidance for dataspace workflows.

- Requirement 3

The platform should be understandable by non-technical users.

- Requirement 4

The platform should support both:

- FIWARE Dataspaces
- Eclipse Dataspaces

- Requirement 5

The platform should support future deployment on EduCloud infrastructure.

Non-Functional Requirements

- User-friendly interface
- Secure storage through redaction
- Deployable on Kubernetes
- Support future FIWARE integration

Functional Requirements

- Execute catalog discovery
- Execute contract negotiation
- Execute transfer initiation
- Retrieve EDR metadata
- Retrieve provider data
- Store execution traces
- Display scenario progress
- Export findings

Development Methodology

The project followed an iterative prototyping approach:

1. Research EDC MVD
2. Build technical trace dashboard
3. Gather stakeholder feedback
4. Redesign UI around use cases
5. Develop Scenario Wizard
6. Implement trace system
7. Prepare EduCloud deployment

System Overview

The platform acts as an orchestration and visualization layer above an existing Eclipse MVD deployment. Architecture:

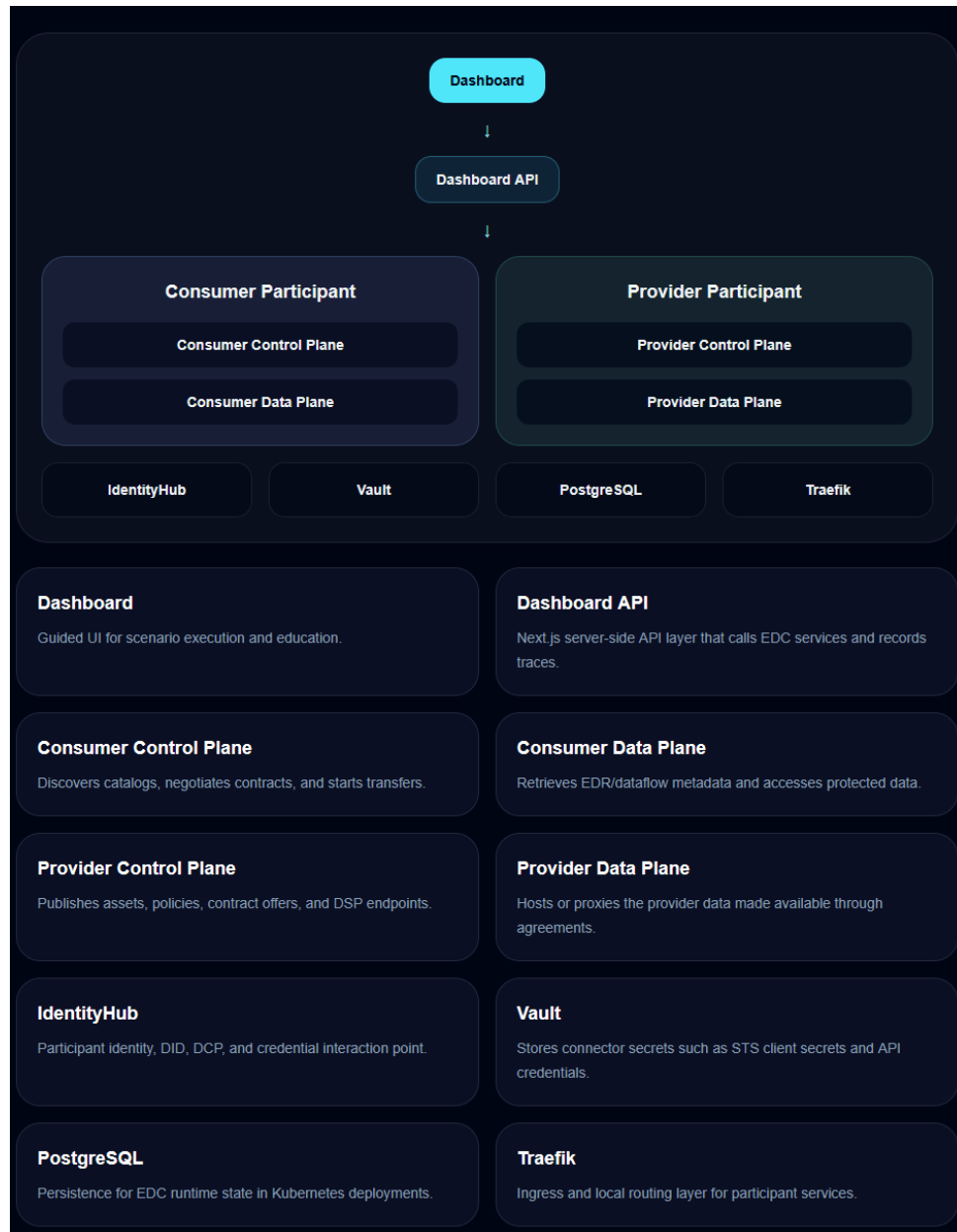


Image 2: /architecture System Architecture of the Dataspace Use-Case Validation Platform

The dashboard does not replace the MVD. Instead, it communicates with the existing MVD deployment through documented APIs and records the resulting interactions. This separation allows the dashboard to remain lightweight while accurately representing real dataspace behavior.

Platform Architecture

	Frontend	Backend	Database
Technology:	• Next.js • React • TypeScript	• Next.js API routes	• SQLite
Responsibilities:	• Scenario Wizard • Trace visualization • Sequence diagrams • Configuration interface • Workflow execution	• Execute MVD requests • Store traces • Redact sensitive information • Coordinate workflow execution	• traces • execution history • dashboard settings
Notes:			The database is stored locally in: data/mvd-traces.sqlite

This project's repository - [Github](#).

Dataspace Scenario Wizard

The Scenario Wizard is the primary stakeholder-facing component.

Its purpose is to transform complex dataspace interactions into understandable workflows.

Users select:

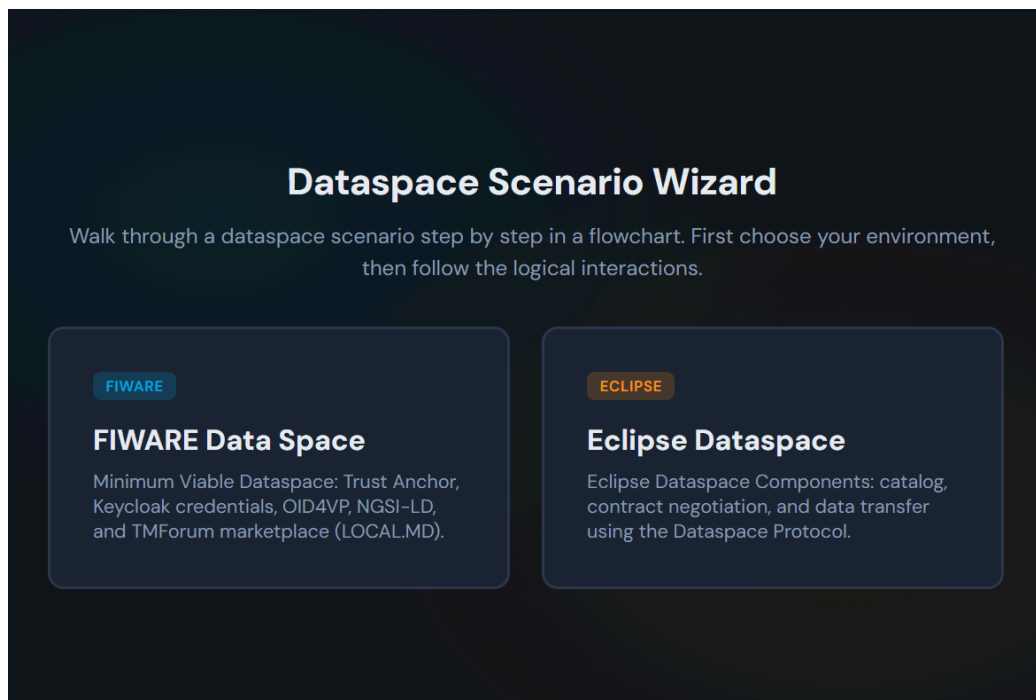
- FIWARE Dataspace
- Eclipse Dataspace

The platform then guides them through the required steps.

Benefits

- educational value
- guided execution
- progress visualization
- stakeholder demonstrations

The wizard was introduced following stakeholder feedback indicating that the original dashboard was too technical and difficult for non-technical users to understand.



*Image 3: Dataspace Scenario Wizard Main Screen suggestion (1) *by stakeholders*

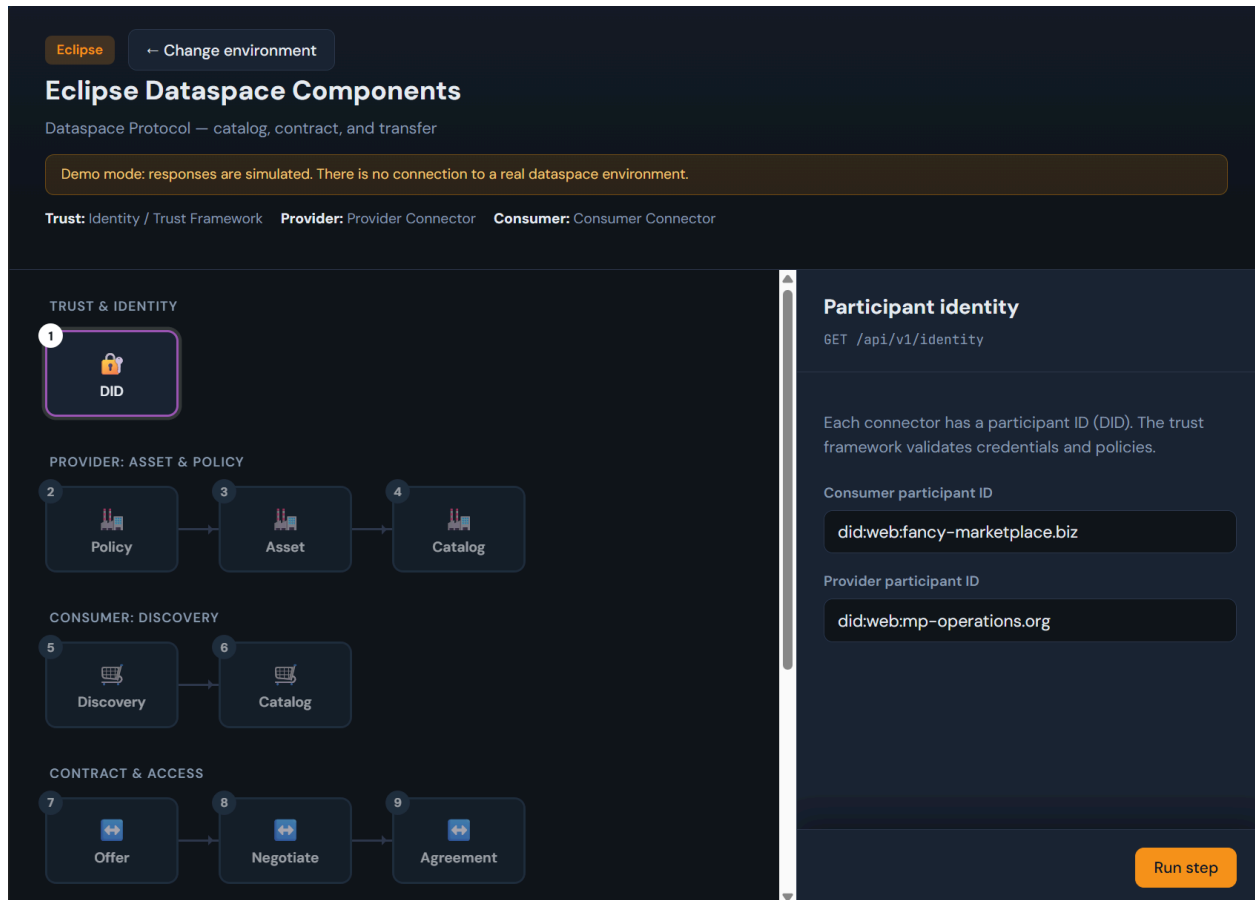


Image 4: Dataspace Scenario Wizard EDC workflow page suggestion (2)

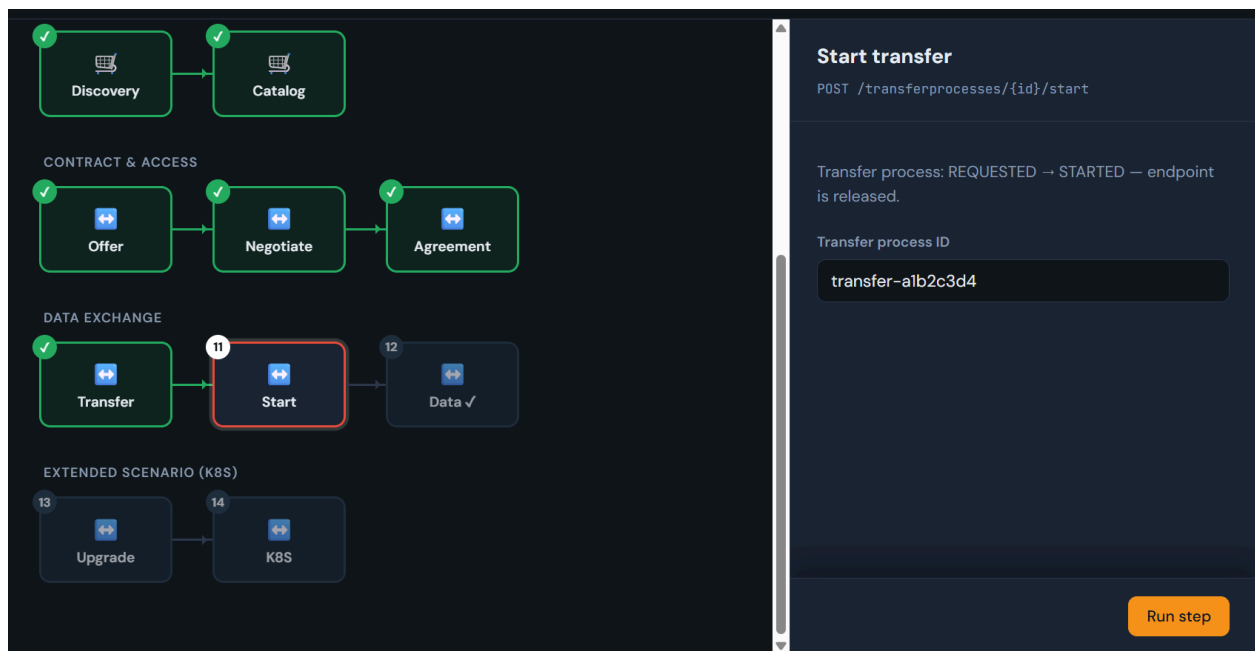


Image 5: Example of progress progression within the UI suggestion (3)

MVD TRACE

EDC Flow Dashboard

Overview

Catalog

Negotiation

Transfer

Data Access

Traces

Settings

Traces the MVD catalog, negotiation, transfer, EDR, and data fetch calls through a local BFF.

N

OVERVIEW

Eclipse EDC MVD trace dashboard

New Trace

Run Full Demo Flow

Catalog Explorer

Requests the provider catalog through the consumer control plane and extracts asset/offer IDs.

Request Catalog

assetId

contractOfferId

Contract Negotiation Monitor

Starts negotiation for the selected offer and polls until the agreement ID is available.

Start Negotiation

Poll Negotiation

contractNegotiationId

contractAgreementId

Transfer Monitor

Starts an HTTP proxy transfer from the agreement and polls transfer state.

Start Transfer

Poll Transfer

transferProcessId

EDR / Data Access

Reads open dataflow metadata from the consumer data-plane proxy, then fetches the final data.

Image 6: EDC Dashboard first UI before feedback

EDC EDUCATION

Dataspace Validation Platform

Guided use-case validation for Eclipse Dataspace Components, with technical traces available on demand.

Overview

Use Cases

Scenario Wizard

Execution History

Architecture

Deployment Status

Advanced Diagnostics

Settings

Scenario mode explains what happened and why. Advanced Diagnostics keeps the EDC protocol details available for deeper review.

DATASPACE SCENARIO WIZARD

Validate a dataspace use case step by step

Run the scenario as a guided educational workflow. The wizard explains what happened and why, while the trace ID is kept for Advanced Diagnostics.

Replay

Run Scenario

UC-E5 - End-to-End Scenario
0% complete

SELECTED USE CASE

UC-E5 · End-to-End Scenario
Ready

Validate the complete dataspace flow.

Success: Identity verification, discovery, negotiation, transfer, and data retrieval all complete successfully.

TRUST

1
Identity Verification
Pending

The platform can reach the configured participant services.

DISCOVERY

2
Catalog Discovery
Pending

A data asset and contract offer are discovered.

POLICY

3
Offer Selection
Pending

The discovered offer is ready for negotiation.

EXCHANGE

4
Contract Negotiation
Pending

A contract negotiation and agreement are created.

POLICY

5
Policy Validation
Pending

The agreement remains valid and is not terminated by policy.

EXCHANGE

6
Transfer Initialization
Pending

A transfer process is created and reaches a started state.

DATA

7
Data Retrieval
Pending

The consumer receives the protected data payload.

DATA

8
Interoperability Findings
Pending

A findings record is created and linked to a trace ID for export or review.

SELECTED STEP

Identity Verification

Confirm that the consumer, provider, and data plane are reachable before starting a protected exchange.

WHAT SUCCESS MEANS

Participants are present and ready for a trusted dataspace interaction.

Run Individual Step

APPLIED SCENARIO DATA

Trace	Created when a step runs
Discovered asset	Waiting for catalog discovery
Offer	Waiting for offer selection
Agreement	Not agreed yet
Transfer	Not started yet

Image 7: /scenario-wizard Final Scenario Wizard Implementation

Eclipse Dataspace Workflow

The platform validates the following EDC workflow:

Step 1 – Catalog Discovery

Consumer requests available provider assets. - Output:

- Asset ID
- Contract Offer ID

Step 2 – Contract Negotiation

Consumer initiates contract negotiation. - Output:

- Negotiation ID
- Agreement ID

Step 3 – Transfer Initiation

Consumer requests transfer. - Output:

- Transfer Process ID

Step 4 – Dataflow / EDR Retrieval

Consumer retrieves endpoint information. - Output:

- EDR Metadata
- Dataflow Information

Step 5 – Data Retrieval

Consumer accesses provider data. - Output:

- Final dataset

Demo from version 1 of the dashboard within this use case - [Video of the Dashboard in execution](#)

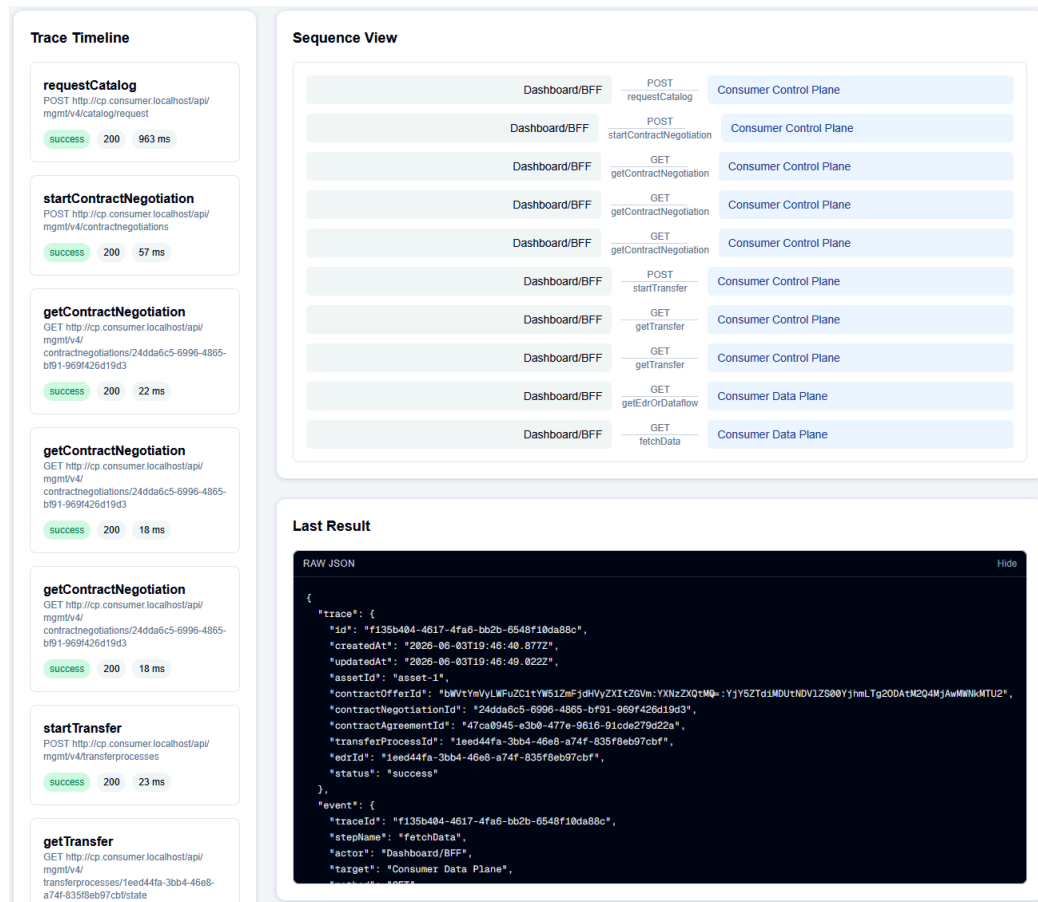


Image 8: (Old) Dashboard after successful run of the test case

Demo from the latest version of the dashboard within the same use case - [Version 2 New UI](#)

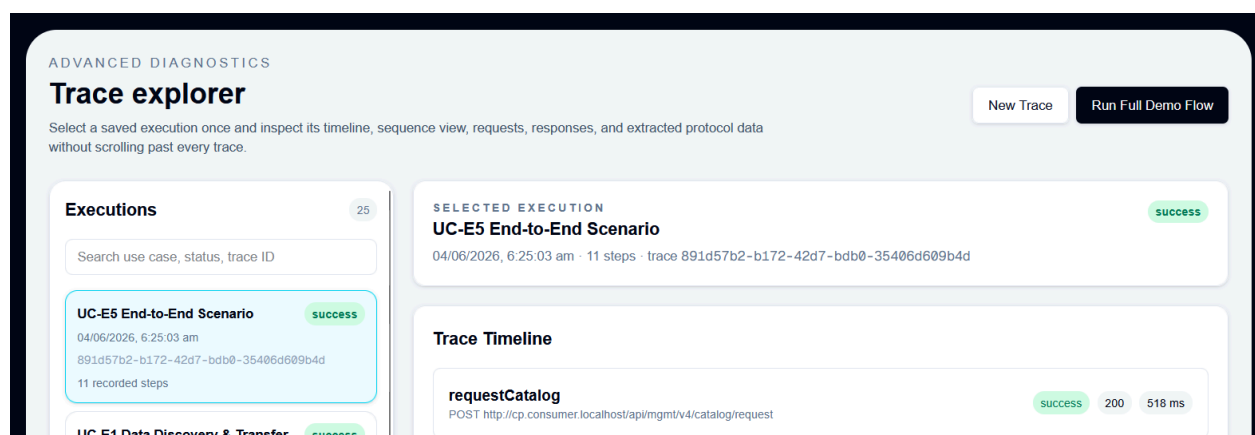


Image 9: /advanced-diagnostics with functioning Full Demo Flow

Trace System

One of the key contributions of the platform is its trace system. Every interaction is recorded.

Examples:

- requestCatalog
- startContractNegotiation
- getContractNegotiation
- startTransfer
- getTransfer
- getEdrOrDataflow
- fetchData

Each trace records:

- request
- response
- duration
- status
- extracted identifiers

Sensitive values are automatically redacted before storage.

EXECUTION HISTORY					
Scenario execution records					
Historical executions are stored locally as traces. The table focuses on scenario outcomes, and the trace ID links each run to Advanced Diagnostics.					
Search by use case, status, or trace ID		All statuses	All use cases	Clear Running	
Timestamp	Use Case	Status	Duration	Trace ID	Action
04/06/2026, 6:25:03 am	UC-E5 End-to-End Scenario local	Success	810 ms	891d57b2-b172-42d7-bdb0-35406d609b4d	Delete
04/06/2026, 3:29:13 am	UC-E1 Data Discovery & Transfer local	Success	161 ms	fae9f959-9d72-48ab-8d80-315f72879263	Delete
04/06/2026, 3:29:10 am	UC-E2 Identity & Trust local	Success	0 ms	32a20859-a308-4c0b-9921-1df30ed2f1e7	Delete
04/06/2026, 3:29:07 am	UC-E3 Policy Enforcement local	Success	132 ms	7d839c6f-56aa-4a3e-9f80-64ebed4711d6	Delete
04/06/2026, 3:29:04 am	UC-E4 Federated Discovery local	Success	88 ms	172cf1cd-5355-46ff-b5f6-f08c5b4e3e3e	Delete
04/06/2026, 3:29:00 am	UC-E6 Interoperability Findings local	Success	0 ms	4e01c79f-7049-4a03-a37e-88109490574d	Delete

Image 10: /execution-history demonstrating the trace system

Use Cases

The platform supports validation of the project use cases.

	UC-E1	UC-E2	UC-E3	UC-E4	UC-E5	UC-E6
	Provider to Consumer Data Transfer	VC Issuance and Verification Future integration	ODRL Policy Enforcement	Federated Catalog Discovery	Full Ecosystem End-to-End Scenario	Cross Framework Interoperability
Validation:	• catalog • negotiation • transfer • retrieval	• DID • VC • Trust Anchor	• policy constraints • permission handling • access control	• provider discovery • contract initiation	Validation of complete workflow.	Integration scenarios.
Success criteria:	The consumer receives provider data.				Successful exchange of findings	

USE CASES

EDC-focused validation scenarios

Each use case is expressed in business terms first. The underlying EDC catalog, negotiation, policy, transfer, and data access calls are still recorded for technical review.

UC-E1
Ready

Data Discovery & Transfer

Runs catalog discovery, contract negotiation, transfer initialization, and final data retrieval.

SUCCESS CRITERIA

Consumer receives the correct data asset.

Run ScenarioView Results

UC-E2
Warning

Identity & Trust

Checks whether participant trust prerequisites are reachable before protected resources are accessed.

SUCCESS CRITERIA

Only valid credential holders can access protected resources.

Run ScenarioView Results

UC-E3
Ready

Policy Enforcement

Shows where policy terms are evaluated before a contract and transfer are allowed.

SUCCESS CRITERIA

Policy engine correctly permits or denies access according to ODRL constraints.

Run ScenarioView Results

UC-E4
Ready

Federated Discovery

Demonstrates discovery of provider assets through dataspace catalog interactions.

SUCCESS CRITERIA

Consumer discovers and negotiates assets through federated catalogs.

Run ScenarioView Results

UC-E5
Ready

End-to-End Scenario

Runs identity verification, discovery, negotiation, policy validation, transfer, and data retrieval.

SUCCESS CRITERIA

Identity verification, discovery, negotiation, transfer, and data retrieval all complete successfully.

Run ScenarioView Results

UC-E6
Warning

Interoperability Findings

Captures findings from execution results, endpoint assumptions, failures, and protocol observations.

SUCCESS CRITERIA

Technical findings are captured and exported.

Run ScenarioView Results

Image 11: /use-cases page with working “Run Scenario” and “View Results” functionalities

Deployment Architecture

Target deployment platform:

EduCloud Kubernetes Cluster

Components:

- Dashboard
- Consumer Connector
- Provider Connector
- Consumer Data Plane
- Provider Data Plane
- IdentityHub
- Vault
- Traefik Gateway

The dashboard container is deployed independently and communicates with the existing MVD services.

EduCloud Readiness

The platform was prepared for EduCloud deployment through:

- Docker containerization
- Kubernetes manifests
- Gateway API support
- Traefik ingress support
- Health endpoints
- Environment-based configuration

This allows deployment to production-like Kubernetes environments.

Challenges Encountered

Major issues encountered during development:

Vault Integration

Transfer processes failed due to missing dataplane credentials.

Investigation required:

- Vault inspection
- Secret validation
- Dataplane key verification

Gateway API Migration

The MVD deployment required Gateway API support and Traefik configuration adjustments.

Data Plane Authorization

Transfer processes occasionally terminated because of credential and token configuration mismatches.

These findings were documented and validated during testing.

Results

The platform successfully demonstrates:

- catalog discovery
- contract negotiation
- transfer initiation
- EDR retrieval
- data access
- trace visualization

The dashboard transforms low-level EDC interactions into a stakeholder-friendly validation environment.

Future Work

Planned improvements:

- Full FIWARE integration
- VC issuance workflows
- Trust Anchor integration
- Cross-framework interoperability testing
- Multi-user support
- Production authentication
- Advanced analytics
- EduCloud production deployment

Conclusion

The Dataspace Use-Case Validation Platform evolved from a debugging dashboard into a stakeholder-oriented validation environment for Eclipse Dataspaces.

The platform improves accessibility, transparency, and educational value while preserving the technical accuracy of the Eclipse MVD workflow.

By combining guided workflows, execution traces, and stakeholder-driven design principles, it provides a practical foundation for validating dataspace use cases across both Eclipse EDC and future FIWARE integrations.

Sources

- <https://github.com/eclipse-edc/Samples>
- <https://github.com/eclipse-edc/Samples/blob/main/transfer/README.md>
- <https://github.com/eclipse-edc/Samples/blob/main/advanced/README.md>
- <https://github.com/eclipse-edc/Samples/blob/main/policy/policy-01-policy-enforcement/README.md>
- <https://dataspace.eclipse.org/>
- <https://www.youtube.com/@eclipsedataspaceconnector9622/videos>
- <https://catena-x.net/>
- <https://eclipse-dataspace-protocol-base.github.io/DataspaceProtocol/2025-1-err1/>
- <https://projects.eclipse.org/working-group/eclipse-dataspace>

Notes